

How Many In-Context Examples Is a Task Description Worth?

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Abstract

While prompts in modern language models typically consist of both a task description and in-context examples, theoretical frameworks for in-context learning (ICL) predominantly focus on the latter. In this work, we adopt a Bayesian perspective on ICL to formally quantify the value of task descriptions. By casting in-context examples as observational data and the task description as a prior, we can measure the description’s value via its prior effective sample size (ESS)—the equivalent number of examples it provides. We analyze this in the setting of in-context linear regression, where Bayes-optimal predictions can be computed exactly. We define the ESS of a description by matching the one-step log-loss regret of a Bayes-optimal predictor using only the description to one using only examples. Our analysis reveals that a description’s worth is governed by two factors: its precision (how much it reduces prior variance) and its reliability (the probability it is correct). We show that for a reliable description, removing a fraction $1 - r$ of the prior variance in a d -dimensional task yields an ESS of approximately $d(1 - r)$. However, for unreliable descriptions, the ESS saturates: the inherent uncertainty imposes a regret floor, so that beyond a certain point increasing the precision of the description yields diminishing returns: at $p = 0.9$, a fiftyfold increase in precision adds a few examples where a reliable description gains a hundred. *[Finally, we investigate whether small meta-trained Transformers mirror these Bayes-optimal ESS values.]* Our claims concern Bayes-optimal predictors and small meta-trained sequence models, not large language models.

1 Introduction

In-context learning (ICL) allows models to adapt to new tasks from a prompt (Brown et al., 2020). In practice, a prompt generally comprises a natural language task description paired with a few input-output demonstrations. However, formal theories of ICL largely abstract prompts as mere collections of demonstrations (Garg et al., 2022; Akyürek et al., 2023; Xie et al., 2022; Genewein et al., 2025). This leaves a significant gap in our understanding, as empirical evidence on the value of task descriptions remains mixed. Some studies show that a good description can be worth hundreds of data points (Le Scao and Rush, 2021; Honda et al., 2025), whereas others find that misleading descriptions perform about as well as informative ones (Webson and Pavlick, 2022) and that enough examples override an explicitly stated bias (Gupta et al., 2025). Even recent theoretical models that incorporate descriptors (Huang and Ge, 2025; Lin et al., 2025; Tong et al., 2026) fail to express the utility of these descriptions on a commensurate scale with the examples themselves.

We quantify the value of a task description as the number of in-context examples it can replace. Following the Bayesian interpretation of ICL (Xie et al., 2022), a sequence model can be seen as performing posterior predictive inference over a latent task. In this view, in-context examples are observations, and the task description acts as a prior. Within Bayesian statistics, the weight of a prior is naturally measured by its *prior effective sample size* (ESS) (Morita et al., 2008). This ESS is primarily driven by two dimensions: the prior’s concentration, and its concordance with the underlying truth (Evans and Moshonov, 2006). Translating this to task descriptions, we identify two key characteristics: **precision** (the extent to which the description reduces posterior uncertainty) and **reliability** (the probability that the description is actually correct).

We compute the ESS as a function of precision and reliability within in-context linear regression (Garg et al., 2022), a setting where the Bayes-optimal predictor is exact. We focus on one-step prediction, mirroring standard prompt usage.

Contributions.

1. We introduce a formal definition for the ESS of a task description by equating the one-step log-loss regret of a description to that of n examples (§3).

2. We establish that for a perfectly reliable description, the ESS scales linearly as $\text{ESS} \approx d(1 - r)$, where d is the dimensionality and r is the precision ratio (§4).
3. We demonstrate that unreliable descriptions induce a hard floor on regret, governed by the description’s error probability $1 - p$ and its entropy $H(p)$. Consequently, the ESS saturates; making an unreliable description infinitely precise yields diminishing returns (§5).
4. *[Pending.]* We design experiments to test whether small, meta-trained Transformers reflect these theoretically derived ESS values (§6).

2 Related Work

Bayesian Perspectives on ICL. ICL can be framed as implicit Bayesian inference (Xie et al., 2022). In the context of linear regression, meta-trained Transformers effectively implement Bayes-optimal ridge regression (Akyürek et al., 2023; Raventós et al., 2023; Panwar et al., 2024). Genewein et al. (2025) evaluate such models using log-loss regret against the Bayes-optimal baseline, a metric we similarly adopt. Yet, these works do not translate the influence of a prior into an equivalent number of in-context examples.

Task Descriptions in ICL Theory. Recent studies have begun exploring task descriptions theoretically. Huang and Ge (2025) include a descriptor token representing the input mean, but because it is independent of the regression weights, its ESS in our framework would be zero. Lin et al. (2025) guide models using hypothesis classes, and Tong et al. (2026) demonstrate how models can exponentially override incorrect instructions given enough examples. Lin and Lee (2024) analyze Gaussian-mixture priors similar to ours, but none of these works quantify the description’s value in terms of sample size.

Effective Sample Size. In statistics, Morita et al. (2008) define ESS via curvature matching, while Reimherr et al. (2021) highlight the need for prediction-based ESS measures. Our approach directly addresses this by defining ESS through predictive regret. Furthermore, our model of reliability draws on robust mixture priors (Schmidli et al., 2014), commonly used in clinical trials to incorporate potentially conflicting historical data.

3 Problem Formulation

Task Distribution. We consider tasks defined by a weight vector $w \in \mathbb{R}^d$. An in-context example consists of an input $x \sim \mathcal{N}(0, I_d)$ and a corresponding label $y = w^\top x + \epsilon$, where the noise is $\epsilon \sim \mathcal{N}(0, \sigma_y^2)$. The base prior over tasks is $w \sim \mathcal{N}(0, s_0^2 I_d)$. For simplicity, we set $\sigma_y = 1$ and define the prior signal-to-noise ratio (SNR) as $a_0 = s_0^2 / \sigma_y^2$.

Task Description Model. We model a task description as a vector $m \in \mathbb{R}^d$ that provides information about w . The description’s *precision ratio* is $r \in (0, 1)$, meaning the variance around the description is $s_\ell^2 = r s_0^2$. The description is generated hierarchically via $m \sim \mathcal{N}(0, (1 - r)s_0^2 I_d)$, ensuring the marginal distribution of w matches the base prior. A description is not always correct. We define its *reliability* p as the probability that the description is accurate. If correct, $w \sim \mathcal{N}(m, s_\ell^2 I_d)$; if incorrect (with probability $1 - p$), w simply follows the base prior $\mathcal{N}(0, s_0^2 I_d)$ independently of m . Consequently, the Bayes-optimal prior conditioned on the description is a robust mixture (Schmidli et al., 2014): $p \mathcal{N}(m, s_\ell^2 I_d) + (1 - p) \mathcal{N}(0, s_0^2 I_d)$. This formulation treats the description as a weighted numerical prior rather than raw text to be parsed.

Regret and Effective Sample Size. We evaluate predictors based on one-step predictive log-loss for a new query x . The regret is the expected difference in log-loss between the predictor and an oracle that knows w exactly, the expectation taken over tasks, inputs, and labels (Genewein et al., 2025).

Definition 1 (Effective Sample Size). *Let R^{desc} be the regret of a Bayes-optimal predictor given only the task description, and $R^{\text{ex}}(n)$ be the regret of a Bayes-optimal predictor under the base prior given n examples. The effective sample size (ESS) of the description is the value n (interpolated continuously) where $R^{\text{ex}}(n) = R^{\text{desc}}$.*

Table 1: Exact ESS of a reliable description ($p = 1$) for $a_0 = 10$, averaged over three seeds (standard deviation across seeds below 2% of each value). The approximation $d(1 - r)$ holds well as long as the exact ESS does not exceed d .

d	r	ESS	$d(1 - r)$
4	0.5	1.77	2.0
4	0.2	3.32	3.2
16	0.5	7.62	8.0
16	0.2	12.6	12.8
16	0.05	16.9	15.2
16	0.01	26.1	15.8
64	0.5	31.6	32.0
64	0.2	50.7	51.2
64	0.05	62.1	60.8
64	0.01	73.5	63.4

Since we compare Bayes-optimal predictors under different conditioning sets, our ESS fundamentally measures information content rather than the specific architecture of a learner.

4 The Value of Precision

We first analyze a perfectly reliable description ($p = 1$). Given n examples $X_n \in \mathbb{R}^{n \times d}$, the posterior covariance under a Gaussian prior with SNR a is $\Sigma = \sigma_y^2(I/a + X_n^\top X_n)^{-1}$. This is independent of the examples' labels and the description m . The expected one-step regret simplifies to $\frac{1}{2}\mathbb{E}\log(1 + x^\top \Sigma x)$. A reliable description equates to $n = 0$ and $a = ra_0$, whereas relying solely on examples uses $a = a_0$.

Proposition 1. *Let ψ be the digamma function. As $a_0 \rightarrow \infty$, for $n < d$,*

$$R^{\text{ex}}(n) = \frac{1}{2} \log a_0 + \frac{1}{2} \log 2 + \frac{1}{2} \psi\left(\frac{d-n}{2}\right) + o(1), \quad R^{\text{desc}} = \frac{1}{2} \log(ra_0) + \frac{1}{2} \log 2 + \frac{1}{2} \psi\left(\frac{d}{2}\right) + o(1),$$

so the high-SNR ESS, with $R^{\text{ex}}(n)$ extended to non-integer n through the digamma function, solves $\psi\left(\frac{d-n}{2}\right) - \psi\left(\frac{d}{2}\right) = \log r$. As $rd \rightarrow \infty$, uniformly in $r \in (0, 1)$, its solution is

$$\text{ESS} = (d - 1)(1 - r) + O\left(\frac{1}{rd}\right).$$

Idea. Regret is half the log of the leftover variance in the prediction. At high SNR each example removes all the variance along the one direction it looks at, so $n < d$ examples leave $d - n$ directions untouched; a description shrinks all d directions by the factor r . Matching the two leftovers gives $d - n \approx rd$. The proof, with the error term, is in Appendix A.

The error term makes the range of validity precise: the approximation needs $rd \gg 1$, that is, the description must leave many more than one example's worth of variance unexplained, which is the condition $\text{ESS} < d$ in the first-order form $\text{ESS} \approx d(1 - r)$. At $a_0 = 100$ the exact ESS is within 0.1 of $(d - 1)(1 - r)$ for $d \in \{16, 64\}$ and $r \in \{0.5, 0.2\}$ (values 7.50, 12.1, 31.6, 50.5 against 7.5, 12.0, 31.5, 50.4).

Table 1 illustrates this alignment. A description eliminating 50% of the prior variance is worth roughly half the dimension in examples. However, once the description is precise enough that its ESS surpasses d , the approximation breaks down. The true ESS scales beyond d because d noisy examples cannot pinpoint w perfectly.

5 The Bottleneck of Reliability

In reality, descriptions can be flawed or ambiguous. When providing an unreliable description without examples, the Bayes-optimal predictive distribution becomes a mixture. Without observations that could identify the correct component, the predictor must average over both.

Table 2: ESS of unreliable descriptions ($d = 16$, $a_0 = 10$), three seeds; standard deviation across seeds at most 1% of each value. As precision increases (smaller r), the ESS is capped by the reliability p .

r	$p = 1$	$p = 0.99$	$p = 0.9$	$p = 0.7$
0.5	7.6	7.5	6.3	4.1
0.05	16.9	16.6	14.8	11.7
0.001	116.5	64.2	23.1	15.6

Proposition 2. Let $R^{\text{desc}}(p, r)$ be the regret of a description with reliability p and precision ratio r , $R_{p=1}^{\text{desc}}(r)$ the same for a reliable one, and $R^{\text{ex}}(0)$ the regret with neither description nor examples. For every $p \in (0, 1)$ and $r \in (0, 1)$,

$$p R_{p=1}^{\text{desc}}(r) + (1 - p) R^{\text{ex}}(0) \leq R^{\text{desc}}(p, r) \leq p R_{p=1}^{\text{desc}}(r) + (1 - p) R^{\text{ex}}(0) + H(p),$$

where $H(p)$ is the binary entropy in nats. In particular $R^{\text{desc}}(p, r) \geq (1 - p) R^{\text{ex}}(0)$ for every r , and as $r \rightarrow 0$ the regret lies between $(1 - p) R^{\text{ex}}(0)$ and $(1 - p) R^{\text{ex}}(0) + H(p)$.

Idea. Compare the learner with a twin who is also told whether the description is correct. The twin’s regret is $p R_{p=1}^{\text{desc}}(r) + (1 - p) R^{\text{ex}}(0)$. The learner cannot beat the twin, since extra information never hurts a Bayes-optimal predictor. And it loses at most $\log(1/p)$ nats when the description is correct and $\log(1/(1 - p))$ when it is wrong, because its predictive puts those weights on the twin’s answer; the average is $H(p)$. The proof is in Appendix B.

The gap between the two bounds is the cost of not knowing whether the description is correct, and it is at most $H(p)$ however precise the description. For $d = 16$ and $a_0 = 10$, $R^{\text{ex}}(0) = 2.51$ nats and $R_{p=1}^{\text{desc}}(0.001) = 0.07$; at $p = 0.9$ the two bounds are 0.32 and 0.64, and the computed regret at $r = 0.001$ is 0.57.

Because $R^{\text{ex}}(n)$ decreases with n , the lower bound caps the ESS at the n where $R^{\text{ex}}(n) = (1 - p) R^{\text{ex}}(0)$, however precise the description. An unreliable description can replace only a bounded number of examples, and the identification cost, up to $H(p)$, lowers the cap further.

Table 2 demonstrates this saturation. For a perfectly reliable description, refining r from 0.05 to 0.001 raises the ESS from 16.9 to 116.5. However, if $p = 0.9$, the ESS only grows from 14.8 to 23.1. Even a 1% unreliability ($p = 0.99$) nearly halves the maximum potential ESS of the most precise description. The same saturation patterns manifest at $d \in \{4, 64\}$ and $a_0 = 100$.

6 Meta-Trained Transformers

[Results pending. The design below is fixed except where marked.]

Design. To verify if empirical models capture these dynamics, we meta-train Transformers on prompts generated via our process in §3. We follow Genewein et al. (2025), evaluating log-loss against the Bayes-optimal baseline on held-out prompts. Prompts are formatted using the prefix embedding method of Huang and Ge (2025): an optional descriptor token containing m , up to 15 example tokens, and a final query token, with two coordinates indicating token types. Because there is no positional encoding, examples remain exchangeable. We employ a standard architecture (Garg et al., 2022): 12 layers, 8 heads, and an embedding dimension of 256, optimized with Adam at a learning rate of 10^{-4} using fresh prompts for every step. *[Output and loss (predictive log-loss as in Genewein et al. (2025) or squared error as in Garg et al. (2022)) and the scoring of prompt positions are to be finalized.]* Crucially, the descriptor only supplies m ; the values of r and p must be implicitly learned by the network during meta-training. We set $d = 5$, $a_0 = 10$, and omit descriptors in 50% of the training data.

Readouts. We evaluate the models one factor at a time: using a precise reliable descriptor ($r = 0.05, p = 1$), evaluating that same model on unreliable descriptors ($p = 0.9$), and using a coarse reliable descriptor ($r = 0.5, p = 1$). We report regret curves with and without descriptions and compute the empirical ESS across three seeds. We will then meta-train a separate model explicitly at $p = 0.9$.

Bayes-Optimal Baselines. Under these settings, the theoretical Bayes-optimal ESS targets are 6.7 for the precise descriptor and 2.2 for the coarse one (using log-loss). At $p = 0.9$, these drop to 5.0 and 1.8, respectively. (Under squared error, the baseline targets are 7.4 and 2.7 for $p = 1$, falling to 4.4 and 2.1 at $p = 0.9$).

[Table: network versus Bayes-optimal ESS and regret curves. Figure: regret against n .]

7 Discussion

Our analysis offers a precise vocabulary for discussing the utility of task descriptions in prompts. For a Bayes-optimal predictor, a task description is worth a specific number of in-context examples. Assuming perfect reliability, this worth scales linearly with the proportion of prior variance the description eliminates. However, reliability acts as a strict bottleneck. Because a predictor must hedge against the possibility that the description is incorrect, the value of very precise descriptions saturates if they are not perfectly reliable.

Limitations and Future Work. Our theoretical claims are restricted to Bayes-optimal predictors and small-scale meta-trained Transformers in a linear-Gaussian setting, chosen to permit exact calculation. Scaling these insights to large language models operating on complex text remains an open challenge. Additionally, our ESS metric applies to one-step prediction. Over long horizons, sequential feedback allows predictors to detect and adapt to incorrect descriptions, meaning ESS will dynamically evolve with the prediction horizon—a compelling avenue for future research. *[Network results pending.]*

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A Proof of Proposition 1

Idea. Regret is half the log of the leftover variance in the prediction. At high SNR each example removes all the variance along the one direction it looks at and none elsewhere, so $n < d$ examples leave $d - n$ directions untouched. A description instead shrinks every one of the d directions by the factor r . The ESS is the n at which the two leftovers match.

Proof. Step 1: regret is leftover variance. The Bayes predictive for y is the true conditional law of y given the data, a Gaussian with variance $1 + x^\top \Sigma x$ where Σ is the posterior covariance of w . The expected log loss of a true conditional law is its entropy, $\frac{1}{2} \log(2\pi e(1 + x^\top \Sigma x))$; the oracle’s predictive is $\mathcal{N}(w^\top x, 1)$ with entropy $\frac{1}{2} \log(2\pi e)$. So the regret is $\frac{1}{2} \mathbb{E} \log(1 + x^\top \Sigma x)$: half the log of the leftover variance.

Step 2: examples. Write $X_n^\top X_n = \sum_{i \leq n} \lambda_i u_i u_i^\top$ with $\lambda_i > 0$ almost surely, and let $P_\perp$ project onto the $d - n$ directions the examples did not look at. Then $\Sigma = (I/a_0 + X_n^\top X_n)^{-1}$ has eigenvalue a_0 on those directions and $1/(1/a_0 + \lambda_i)$ on the others, so

$$1 + x^\top \Sigma x = a_0 \|P_\perp x\|^2 + T, \quad T = 1 + \sum_{i \leq n} \frac{(u_i^\top x)^2}{1/a_0 + \lambda_i}.$$

Because x is a fresh isotropic Gaussian, $\|P_\perp x\|^2 \sim \chi_{d-n}^2$, independent of X_n . Hence

$$R^{\text{ex}}(n) = \frac{1}{2} \log a_0 + \frac{1}{2} \mathbb{E} \log \chi_{d-n}^2 + \frac{1}{2} \mathbb{E} \log \left(1 + \frac{T/a_0}{\|P_\perp x\|^2} \right).$$

The last term is nonnegative, and $T/a_0 = 1/a_0 + \sum_{i \leq n} (u_i^\top x)^2 / (1 + a_0 \lambda_i)$ decreases to 0 as $a_0 \rightarrow \infty$, so by monotone convergence the term is $o(1)$. (It is finite at any finite a_0 because $R^{\text{ex}}(n)$ is.)

Step 3: description. Here $n = 0$ and $\Sigma = r a_0 I$, so $1 + x^\top \Sigma x = 1 + r a_0 \|x\|^2$ with $\|x\|^2 \sim \chi_d^2$, and the same argument gives $R^{\text{desc}} = \frac{1}{2} \log(r a_0) + \frac{1}{2} \mathbb{E} \log \chi_d^2 + o(1)$.

Step 4: match them. The $\log a_0$ terms cancel, and $\mathbb{E} \log \chi_k^2 = \psi(k/2) + \log 2$. Taking $\psi((d-n)/2)$ as the extension of $R^{\text{ex}}(n)$ to non-integer n , the high-SNR ESS solves

$$\psi\left(\frac{d-n}{2}\right) - \psi\left(\frac{d}{2}\right) = \log r,$$

which has one root in $(0, d)$ because ψ is increasing and $\log r < 0$. Read it as: $d - n$ untouched directions must be as big, on a log scale, as d directions each shrunk by r . Replacing $\mathbb{E} \log \chi_k^2$ by $\log k$ gives the first-order answer $d - n = rd$, that is $\text{ESS} \approx d(1 - r)$.

Step 5: the error term. Let $k = d - n$. As $rd \rightarrow \infty$, $\psi(k/2) = \psi(d/2) + \log r \rightarrow \infty$ forces $k \rightarrow \infty$. The log of a χ_k^2 variable averages a little below the log of its mean: $\psi(z) = \log z - \frac{1}{2z} + O(z^{-2})$ as $z \rightarrow \infty$. Hence

$$\log \frac{k}{d} - \frac{1}{k} + \frac{1}{d} + O(k^{-2}) = \log r \implies k = rd \left(1 + \frac{1}{k} - \frac{1}{d} + O(k^{-2}) \right).$$

So $k = rd(1 + o(1))$ and $rd/k = 1 + O(1/k)$. Multiplying out, $k = rd + 1 - r + O(1/(rd))$, and $\text{ESS} = d - k = (d - 1)(1 - r) + O(1/(rd))$. The correction $-(1 - r)$ comes from the $-1/k + 1/d$ term: the $d - n$ untouched directions are fewer, so their chi-square sits further below its mean on the log scale. $\square$

B Proof of Proposition 2

Idea. Compare the learner with a twin who is also told whether the description is correct. The twin's regret is easy to write down. The learner cannot beat the twin, because extra information never hurts a Bayes-optimal predictor. And the learner cannot be much worse: its predictive puts weight p on the twin's answer when the description is correct and $1 - p$ when it is wrong, so it loses at most $\log(1/p)$ or $\log(1/(1 - p))$ nats against the twin, which averages to $H(p)$.

Proof. Let $Z = 1$ if the description is correct and $Z = 0$ otherwise.

Step 1: the predictive is a p -mixture. Seeing m says nothing about Z , because m has the same law $\mathcal{N}(0, (1 - r)a_0 I)$ either way, and x is independent of Z . With no examples there is nothing else to update on, so the posterior weight on $Z = 1$ stays p and the Bayes predictive is

$$q(y) = p f_1(y) + (1 - p) f_0(y), \quad f_1 = \mathcal{N}(m^\top x, 1 + ra_0 \|x\|^2), \quad f_0 = \mathcal{N}(0, 1 + a_0 \|x\|^2),$$

where f_1 is the predictive given a correct description and f_0 the predictive under the base prior.

Step 2: the twin. Given Z , the true conditional density of y is f_Z . So the twin predicts with f_1 when $Z = 1$, with regret $R_{p=1}^{\text{desc}}(r)$, and with f_0 when $Z = 0$, with regret $R^{\text{ex}}(0)$. Its regret is $p R_{p=1}^{\text{desc}}(r) + (1 - p) R^{\text{ex}}(0)$.

Step 3: the learner cannot beat the twin (lower bound). Given (x, m, Z) , the expected log loss of any density q exceeds that of the true density f_Z by $\text{KL}(f_Z \| q) \geq 0$. Averaging over Z gives $R^{\text{desc}}(p, r) \geq p R_{p=1}^{\text{desc}}(r) + (1 - p) R^{\text{ex}}(0)$.

Step 4: the learner is at most $H(p)$ worse (upper bound). Pointwise, $q \geq p f_1$ and $q \geq (1 - p) f_0$. So when $Z = 1$, $-\log q \leq -\log f_1 + \log(1/p)$, and when $Z = 0$, $-\log q \leq -\log f_0 + \log(1/(1 - p))$. Averaging, the learner's regret exceeds the twin's by at most $p \log(1/p) + (1 - p) \log(1/(1 - p)) = H(p)$.

Step 5: the limits. $R_{p=1}^{\text{desc}}(r) = \frac{1}{2} \mathbb{E} \log(1 + ra_0 \|x\|^2)$ is nonnegative, which gives the floor $(1 - p) R^{\text{ex}}(0)$ for every r , and it decreases to 0 as $r \rightarrow 0$ by monotone convergence, which gives the two limits. $\square$